Project Design Phase-II
Data Flow Diagram & User Stories

Date	12 October 2026
Team ID	SWUID2026403439
Project Name	TixBox – Online Movie, Event & Entertainment Ticket Booking System
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) illustrates how data moves within the TixBox ticket booking platform. It shows how users (customers, cinema managers, event organizers and admins) and external systems (mock payment gateway and SMTP email server) interact with the system, how information flows between processes, and where the data is stored. The diagrams below follow the SRS and are drawn at two levels: Level 0 (context) and Level 1 (major processes).

Legend:

Symbol	Meaning
Rectangle	External entity (user or external system)
Circle	Process that transforms data
Open-ended bar (D1–D9)	Data store (database table group)
Arrow	Data flow, labelled with the data moved

DFD Level 0 – Context Diagram:

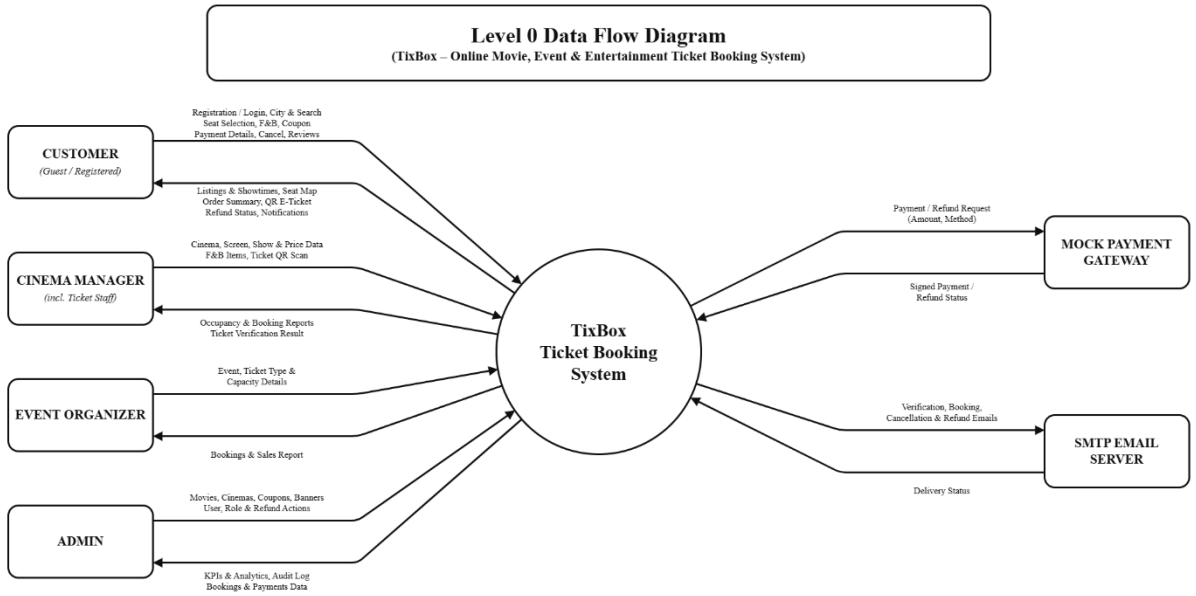


Figure 1: DFD Level 0 – TixBox

DFD Level 1 – Major Processes:

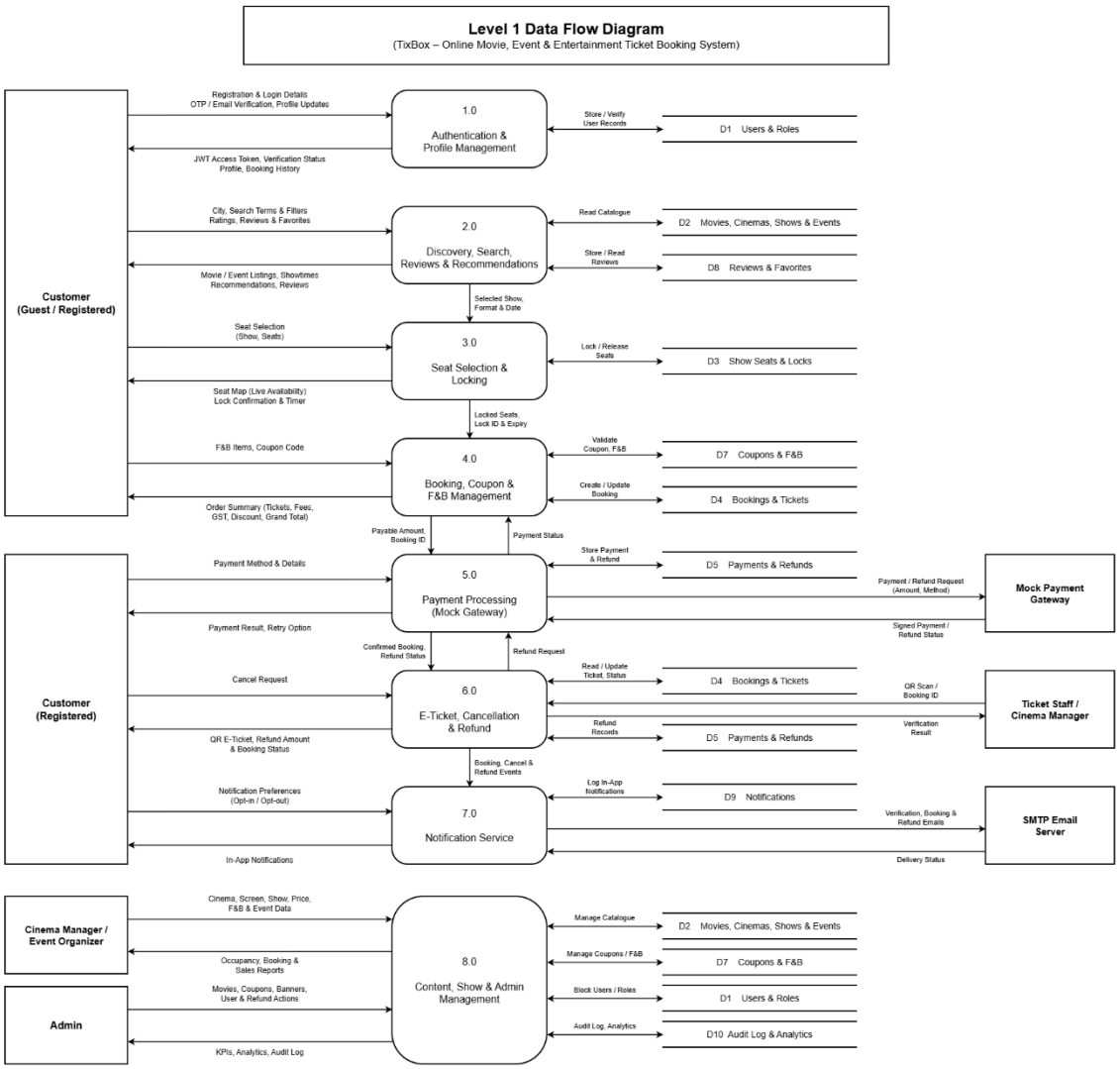


Figure 2: DFD Level 1 – TixBox (processes 1.0 to 10.0, data stores D1 to D9)

Level 1 Process Description:

No.	Process	Description	Data Stores
1.0	Authentication & Profile	Registration, OTP/email verification, login, JWT issue, password reset, profile and RBAC	D1
2.0	Discovery & Search	City selection, home sections, movie/cinema/event details, search, filter, sort	D2, D6
3.0	Seat Selection & Locking	Seat map, atomic seat lock with 10-minute expiry, auto-release	D3
4.0	Booking Management	Order summary with F&B, coupon, fees and tax; booking status handling	D3, D4, D7
5.0	Payment Processing	Mock gateway initiate, verify, refund; signature check; server-side amount	D4, D5
6.0	E-Ticket, Cancel & Refund	Signed QR ticket, staff verification, cancellation, refund slabs	D3, D4, D5
7.0	Content Management	Movies, cinemas, screens, shows, events, F&B, coupons, banners	D2, D3, D6, D7
8.0	Reviews & Recommendations	Ratings, reviews, favorites, personalised recommendations	D4, D8
9.0	Notification Service	In-app notifications and asynchronous emails with retry	D9
10.0	Admin & Analytics	KPIs, user management, reports, audit log	D1, D4, D5, D9

User Stories

User Story Table – TixBox

User Type	Functional Requirement (Epic)	User Story / Task	Acceptance Criteria	Priority	Release
Customer	Authentication	As a guest, I can register and verify my account by email or OTP so that I can book tickets.	Account is created; unverified users cannot book.	High	Sprint-1
Customer	Discovery	As a customer, I can select my city and search or filter movies and events.	Results match city, language, genre and format filters.	High	Sprint-1
Customer	Seat Selection	As a customer, I can pick seats on a live seat map and hold them temporarily.	Seats are locked for 10 minutes with a countdown; no double booking.	High	Sprint-2
Customer	Booking	As a customer, I can add F&B and a coupon and see a full order summary.	Total shows tickets, fee, GST, F&B, discount and grand total.	High	Sprint-2
Customer	Payment	As a customer, I can pay using the mock gateway (UPI, card, net banking, wallet).	Success confirms the booking; failure releases the seats.	High	Sprint-2
Customer	E-Ticket	As a customer, I can get a QR e-ticket and download it as PDF.	QR ticket appears in My Bookings and is emailed.	High	Sprint-2
Customer	Cancellation	As a customer, I can cancel a booking before the deadline and see my refund.	Refund breakup is shown; booking becomes Cancelled then Refunded.	Medium	Sprint-3
Cinema Manager	Content Management	As a cinema manager, I can create shows and set ticket prices for my cinema.	Overlapping shows are rejected; shows appear to customers.	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story / Task	Acceptance Criteria	Priority	Release
Cinema Manager	Ticket Verification	As ticket staff, I can scan a QR code to verify a ticket.	Valid ticket is accepted once; reuse is rejected.	Medium	Sprint-3
Event Organizer	Events	As an organizer, I can publish an event with ticket categories and capacity.	Event is listed; sold tickets never exceed capacity.	Medium	Sprint-3
Customer	Reviews	As a customer, I can rate and review a movie I booked.	Review appears with updated average rating.	Medium	Sprint-3
Admin	Administration	As an admin, I can manage users, coupons and view analytics and the audit log.	Actions are logged; dashboard shows KPIs and charts.	High	Sprint-3